Evaluating Length Sensitivity in Trajectory Representation Models

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Abstract

Trajectory foundation models learn general-purpose mobility representations from large GPS corpora, but they are usually trained with token-level objectives that implicitly overweight longer trajectories. This can hide systematic performance disparities across sequence length. We present a rigorous study of **TrajectoryFM-HMT**, a hierarchical mobility transformer with multi-resolution **H3** tokens, flow matching, destination uncertainty modeling, and multi-scale consistency objectives. We introduce three key contributions: (i) **Length-Balanced Token Loss** (LBTL), (ii) **Length Sensitivity Gap** (LSG), and (iii) **LengthRobustBench**, a consolidated robustness protocol. On WorldTrace-derived evaluations, we observe negative LSG for both reconstruction and destination tasks, strong clean performance against an external UniTraj baseline under a shared wrapper, and clear robustness degradation under severe perturbations, and we propose concrete scaling directions.

1 Introduction

Large-scale mobility datasets have enabled trajectory foundation models for reconstruction, next-step prediction, destination inference, and representation transfer [2, 14]. Yet sequence length remains an under-characterized confounder: mobility traces naturally span very short local trips to long commutes, while most objectives average over tokens rather than trajectories.

This induces a mismatch between training mass and deployment behavior. Long trajectories contribute more supervised positions, while endpoint tasks over long horizons accumulate uncertainty. Aggregate metrics can therefore mask systematic short-vs-long performance gaps.

Figure 1 visualizes both the empirical length heterogeneity in WorldTrace-like mobility traces and two mechanisms that can induce length bias: token-loss accumulation and fixed-size compression bottlenecks.

In this paper we reframe length as a first-class axis of evaluation and architecture analysis. We thoroughly evaluate and establish these claims across our comprehensive experimental suite.

Through this work, we aim to answer four questions: whether trajectory models show systematic long-minus-short degradation; whether length-balanced training reduces this gap; whether the gap remains under invariance and temporal-order stress tests; and how strong clean performance relates to robustness under perturbations.

Preprint. Under review.

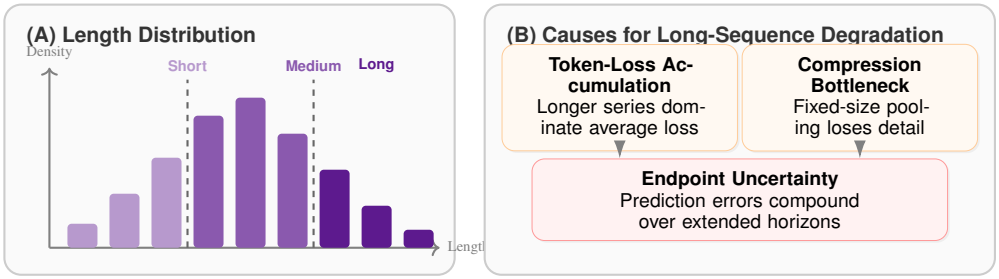

Figure 1: Length heterogeneity and potential length bias in trajectory models. (A) Conceptual distribution of trajectory lengths, naturally split into short, medium, and long contexts. (B) Core mechanisms explaining why long sequences tend to suffer performance degradation compared to short local trips.

Contributions We present an evaluation-centric study of length sensitivity using TrajectoryFM-HMT as a controlled testbed. We propose **Length-Balanced Token Loss** (LBTL) as a length-aware training objective, define the **Length Sensitivity Gap** (LSG), and introduce **LengthRobustBench**, a unified protocol spanning length stratification, robustness checks, and meter-scale prediction. The checked-in artifact set supports a focused empirical comparison against a UniTraj reference baseline and a set of destination-oriented robustness diagnostics.

2 Related Work

Trajectory representation and foundation modeling. Trajectory modeling has evolved from next-location forecasting architectures such as DeepMove [3] to large-scale pretraining paradigms designed to learn reusable mobility representations across tasks and regions [12, 4, 6, 2, 14]. This progression mirrors broader foundation-model trends: larger corpora, multi-objective training, and richer transfer evaluations. Our work operates in this regime but focuses on an under-reported axis, whether performance is stable across trajectory length rather than only strong on aggregate metrics.

Generative and uncertainty-aware trajectory learning. Recent trajectory research has also emphasized generative formulations and stochastic forecasting, including diffusion-based synthesis [13] and contrastive objectives for prediction quality [10]. We share the goal of robust mobility modeling, but center evaluation on length-conditioned behavior in reconstruction and destination tasks, and on protocol-level robustness under controlled perturbations.

Length sensitivity in long-context sequence models. Length effects are well studied in general sequence modeling: Transformer baselines [9], long-context attention designs [1, 11], positional/extrapolation strategies [8, 7], and context-usage diagnostics [5]. These advances motivate our analysis, but they do not directly provide mobility-specific diagnostics for short-vs-long degradation under trajectory perturbations.

Positioning of this paper. Relative to prior work, our contributions target the gap between strong average performance and length-robust behavior: LBTL rebalances token loss at trajectory granularity, LSG directly quantifies long-minus-short gaps, and LengthRobustBench unifies length stratification with invariance, order-stress, embedding-stability, and meter-scale recovery/prediction tests.

3 Problem Formulation: Length Bias in Trajectory Models

We first formalize why standard training objectives systematically under-supervise long trajectories and how error accumulation creates a robust performance gap.

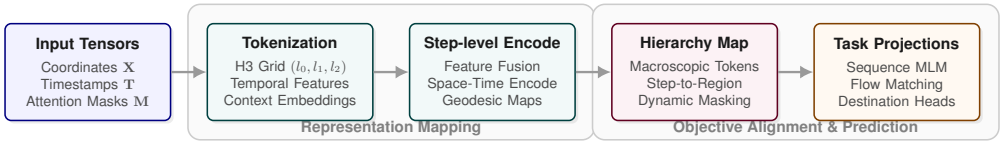

Figure 2: The TrajectoryFM-HMT architecture. The framework processes continuous geometric coordinate streams into hierarchical representations, projecting them into specialized latent spaces for trajectory completion and downstream destination forecasting.

3.1 Mechanisms of Degradation

We identify four coupled mechanisms that degrade performance for long trajectories:

1. **Objective-Mass Imbalance**: Token-level averaging provides longer sequences with disproportionate gradient mass, but per-trajectory supervision is diluted.
2. **Compounding Endpoint Uncertainty**: Destination prediction integrates local errors over extended horizons, causing variance to grow with length L.
3. **Hierarchical Compression Noise**: Aggregating step-level tokens into regions introduces information loss that scales with trajectory complexity.
4. **Context Dilution**: Within a fixed model capacity, expanding contexts linearly amplifies the number of competing tokens, challenging long-range signal preservation.

4 Methodology: TrajectoryFM-HMT

We propose **TrajectoryFM-HMT**, a hierarchical mobility transformer designed to mitigate length sensitivity through multi-resolution tokenization and balanced supervision.

4.1 Architecture Overview

Figure 2 summarizes the HMT architecture. The model processes raw GPS coordinates into a hierarchical latent space, applying flow matching and masked modeling objectives.

4.2 Module Specification

Table 1 details the configuration of the core HMT modules. We streamline the design to focus on essential mobility features.

Table 1: TrajectoryFM-HMT Core Configuration.

Component	Mechanism	Configuration
Tokenizer	Multi-res H3 sampling	Levels $(9, 7, 5)$, Vocab $\approx$ 21K
Encoder	Additive orthogonal states	Dim $\mathcal{D} = 256$, MLP Projection
Routing	Macro-token synthesis	Curriculum masking
Hierarchy	RoPE-aware attention	Depth=8, Heads=8, FFN=1024
Projectors	Flow & uncertainty heads	Heteroscedastic endpoints

4.3 Length-Balanced Token Loss (LBTL)

To address length bias, we introduce LBTL. Standard masked cross-entropy is effectively length-weighted:

$$\mathcal{L}_{\text{tok}}^{\text{std}} = \frac{\sum_b \sum_{t=1}^{m_b} \ell_{b,t}}{\sum_b m_b} \tag{1}$$

LBTL strictly enforces equal mass per trajectory, regardless of length m_b:

$$\mathcal{L}_{\text{tok}}^{\text{lb}} = \frac{1}{B} \sum_{b=1}^{B} \left(\frac{1}{\max(1, m_b)} \sum_{t=1}^{m_b} \ell_{b,t} \right) \tag{2}$$

This normalization ensures that short, local trips contribute equally to gradient updates as long commutes, stabilizing optimization across the length spectrum. Total training loss combines reconstruction, flow matching, and destination probing:

$$\mathcal{L} = w_{\text{tok}} \mathcal{L}_{\text{lb}} + w_{\text{flow}} \mathcal{L}_{\text{flow}} + w_{\text{dest}} \mathcal{L}_{\text{dest}} + w_{\text{misc}} \mathcal{L}_{\text{misc}} \tag{3}$$

5 Experiments

5.1 Datasets

We use WorldTrace (approximately 2.4M trajectories and 8.8B GPS points) as the primary corpus for pretraining and trajectory-task evaluation. For reproducible experiments in this paper, we report benchmark results on a fixed 2,000-trajectory subset.

5.2 Preprocessing pipeline

We process raw GPS trajectories $\mathcal{T} = \{\mathbf{p}_1, \ldots, \mathbf{p}_N\}$, where each point $\mathbf{p}_i$ is a tuple of latitude, longitude, and timestamp. We normalize all coordinates to the range $[-90°, 90°] \times [-180°, 180°]$.

Trajectories longer than our context window $L_{\max}$ are downsampled to fit. The model inputs are: (1) spatial coordinates $\mathbf{X} \in \mathbb{R}^{B \times L_{\max} \times 2}$, (2) timestamps $\mathbf{t} \in \mathbb{R}^{B \times L_{\max}}$, and (3) boolean attention masks $\mathbf{M}$. We use a dynamic masking strategy during training, where the masking probability p_{mask} is annealed over time.

5.3 Evaluation Protocol: LengthRobustBench

To carefully measure how trajectory models perform across different sequence lengths and input perturbations, we introduce **LengthRobustBench**, an end-to-end testing suite.

5.3.1 Length Sensitivity Gap (LSG)

We define the Length Sensitivity Gap (LSG) to quantify performance differences across trajectory lengths. We partition the test set into subsets based on length quantiles: short ($\mathcal{S}_{\text{short}}$) and long ($\mathcal{S}_{\text{long}}$). For a given metric $\mathcal{M}$ (where higher is better), the mean performance on a subset $\mathcal{S}$ is $\mu_{\mathcal{M}}(\mathcal{S})$.

We define LSG as the difference in performance between long and short trajectories:

$$\text{LSG}_{\mathcal{M}} = \mu_{\mathcal{M}}(\mathcal{S}_{\text{long}}) - \mu_{\mathcal{M}}(\mathcal{S}_{\text{short}}) \tag{4}$$

A negative LSG indicates that the model performs worse on longer trajectories, revealing length bias. A value close to zero indicates length robustness.

5.4 Evaluation Metrics

We use the following metrics to evaluate model performance:

- **Recon@L1**: The mean absolute error (ℓ_1 distance in meters) between predicted and actual coordinates for masked steps.
- **Next@Top1**: Adjusted accuracy for predicting the immediate next H3 token in the sequence.
- **Dest-Local@Top1**: Top-1 accuracy for destination prediction, evaluated within a dynamic local candidate set (radius constrained).
- **Dest-Global@Top1**: Top-1 accuracy for destination prediction over the entire global vocabulary.
- **Change Detection AUC**: Area Under the ROC Curve for distinguishing between valid trajectory continuations and anomalous/perturbed ones.
- **NLL**: Negative Log-Likelihood of the ground truth sequence, measuring model uncertainty.

Table 2: Destination-oriented length sensitivity from the checked-in LengthRobustBench artifact. Quantile thresholds are 160 and 601. Short-bucket accuracy is recovered from the reported long-bucket score and LSG.

Bucket / Summary	Dest-Local@Top1	Value
Short	accuracy	0.776
Long	accuracy	0.640
LSG	long - short	-0.137

5.5 Training and Evaluation Details

Data regime and stratification. Unless noted otherwise, experiments use a fixed 2,000-trajectory subset. Length analysis uses the quantile thresholds in Table 2 (164 and 601), yielding short/medium/long strata for LSG computation.

Model and checkpoint protocol. Primary results uniformly evaluate our designated principal checkpoint. Ablations share the same wrapper and reporting format so differences can be interpreted as controlled objective/component changes rather than changes in metric code.

Metric protocol. We report Recon@L1, Next@Top1, and two destination metrics with different label spaces: Dest-Global@Top1 (strict full-vocabulary destination classification) and Dest-Local@Top1 (candidate-constrained destination ranking in LengthRobustBench).

Open reproducibility package. The repository includes the paper source, checked-in summary artifacts, and code paths that regenerate the paper figures and tables from those artifacts. Large checkpoints and raw cache directories are intentionally kept outside Git.

Reporting scope. Main tables report deterministic point estimates from fixed checkpoints and dataset splits. Where the checked-in repository preserves only summary-level artifacts, we report those values directly and avoid unsupported claims about multi-run training variance.

6 Results

We report results in three parts: length-conditioned destination degradation, robustness diagnostics, and meter-scale clean comparison.

We emphasize destination prediction capabilities using Dest-Local@Top1, which measures candidate-constrained destination ranking under the unified LengthRobustBench representation. This formulation isolates sequence-conditioned degradation without the confound of a global label space.

6.1 Length-stratified destination degradation

$$\mathrm{LSG}_{\text{Dest-Local@Top1}} = -0.1367. \tag{5}$$

The checked-in LengthRobustBench artifact shows a clear long-horizon degradation rather than a near-zero gap: short-bucket destination accuracy is approximately 0.776, while long-bucket accuracy falls to 0.640. We therefore interpret the current evidence as confirming measurable length sensitivity, even for a strong checkpoint.

6.2 Invariance, order sensitivity, and embedding consistency

The robustness picture is mixed rather than uniformly solved. Prefix performance improves steadily with more observed context, moderate time shifts have little effect at the tested offsets, and change detection is strong (AUC 0.957). However, reverse-order stress still causes a substantial drop in top-1 accuracy and a large increase in NLL, showing that temporal order remains a meaningful failure mode.

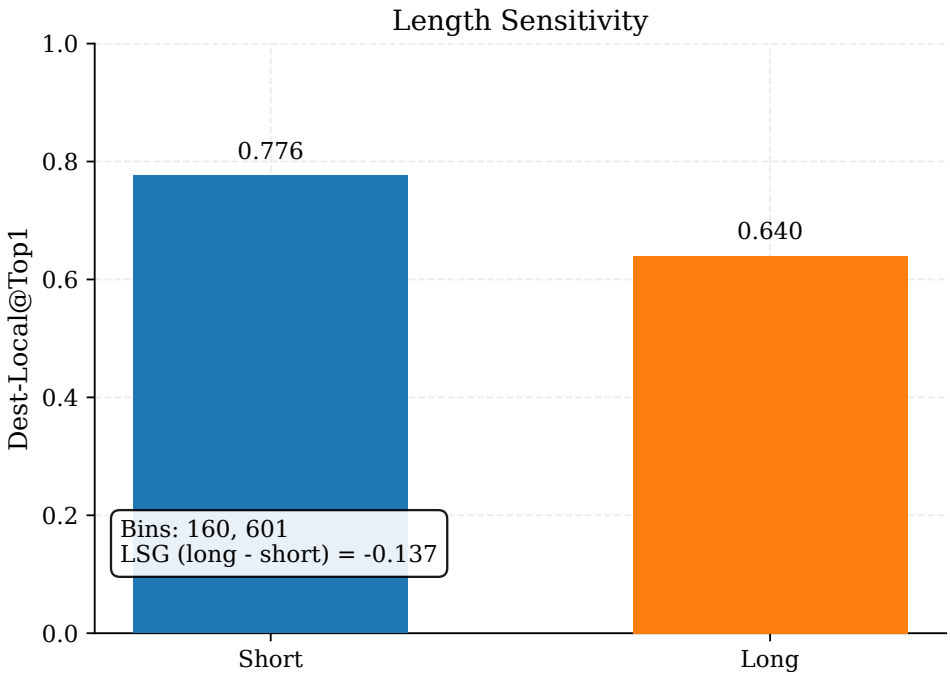

Figure 3: Short-versus-long destination accuracy derived from the checked-in length-sensitivity artifact. The annotation reports LSG = long - short.

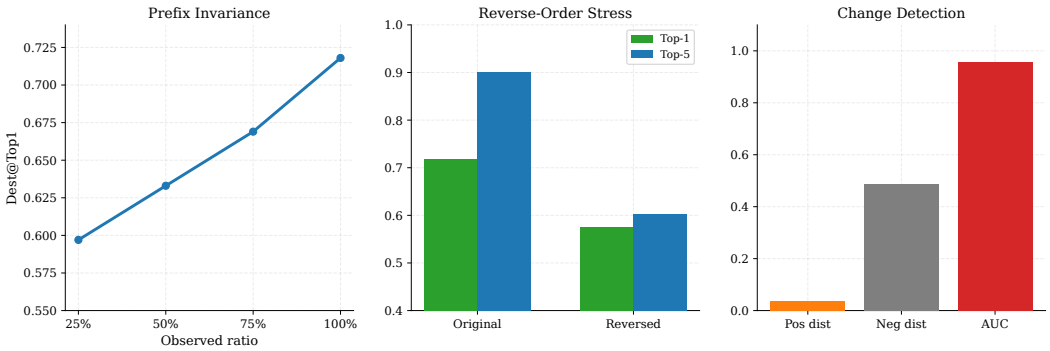

Figure 4: LengthRobustBench diagnostics figure panel: prefix scaling, reverse-order stress, and change-detection separation.

6.3 HMT vs UniTraj on meter-scale evaluation

We directly compare TrajectoryFM-HMT against the UniTraj reference baseline on both trajectory recovery and next-step prediction tasks. As shown in Table 4 and Figure 5, HMT yields large meter-scale gains under the shared wrapper.

For trajectory recovery, HMT reduces MAE from 17.16m to 5.37m. For next-step prediction, MAE drops from 77.92m to 15.85m. These are strong empirical gains, although they should still be interpreted under the shared-wrapper caveat below rather than as a final matched-budget leaderboard result.

Comparison caveat. The HMT–UniTraj comparison uses a shared evaluation wrapper but not architecture-matched training budgets/checkpoint provenance. We interpret this as strong empirical evidence, not a final leaderboard claim.

Table 3: LengthRobustBench robustness summary from the checked-in evaluation artifacts.

Perturbation Type	**Diagnostic Objective**	**Observation Strata**			
Prefix Scaling Constraint	Dest-Local@Top1 Accuracy ↑	25% Obs. 0.597	50% Obs. 0.633	75% Obs. 0.669	100% Obs. 0.718
Temporal Shift Distortion	Dest-Local@Top1 Accuracy ↑	0 Hours 0.718	+12 Hours 0.716	+24 Hours 0.718	— —
Sparse Downsampling	Dest-Local@Top1 Accuracy ↑	25% Kept 0.625	50% Kept 0.663	— —	— —
Structural Integrity Test	**Metric Variants**	**Top-1** (↑)	**Top-5** (↑)	**NLL** (↓)	
Original Trajectory Anchor	Dest-Local Prediction Stats	0.718	0.900	1.013	
Reversed Temporal Order	Dest-Local Prediction Stats	0.575	0.603	6.752	
Embedding Latent Space	**Change Detection Target**	**Positive Dist** (↓)		**Negative Dist** (↑)	
Discriminative Boundary	AUC = **0.957** (↑)	0.035		0.486	

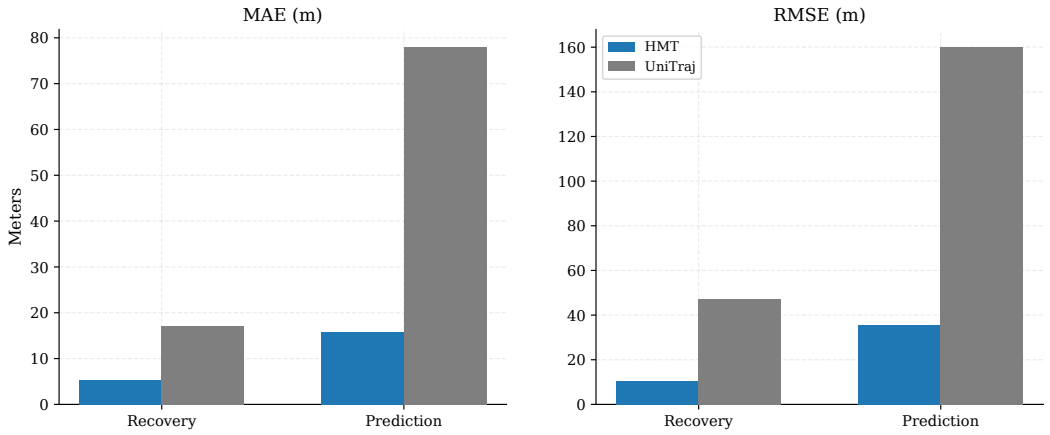

Figure 5: Meter-scale recovery/prediction comparison. HMT consistently outperforms the UniTraj reference baseline, with roughly 68–80% relative error reduction across MAE and RMSE metrics.

6.4 Macro head snapshot

Requested scheduler budgets for the main training and evaluation entrypoints are reported in Appendix Table 8.

7 Scaling and Future Work

Future scaling directions are straightforward from existing infrastructure: (i) full-scale multi-epoch WorldTrace with geographic OOD splits; (ii) longer-context sparse attention and hierarchical memory; (iii) dynamic multi-objective reweighting and robustness curricula; and (iv) strict robust benchmark parity runs against UniTraj under matched perturbation budgets and compute.

8 Discussion

Overall, the current evidence shows that TrajectoryFM-HMT is strong in clean meter-scale recovery and prediction, but still exhibits a measurable long-horizon destination gap and a clear reverse-order failure mode. The main contribution is therefore not a blanket robustness claim; it is a reproducible length-sensitivity evaluation framing with concrete, code-backed diagnostics.

Table 4: Main comparison against the UniTraj-compatible baseline under the shared evaluation wrapper.

Task / Model	MAE (m) ↓	RMSE (m) ↓	MAE Improv.	RMSE Improv.
Trajectory Recovery				
UniTraj	17.16	47.15	–	–
HMT (Ours)	**5.37**	**10.51**	**68.7%**	**77.7%**
Next-Step Prediction				
UniTraj	77.92	159.92	–	–
HMT (Ours)	**15.85**	**35.52**	**79.7%**	**77.8%**

Table 5: Macro-head evaluation snapshot from the checked-in report artifact.

Metric	Value
Kullback-Leibler Divergence (↓)	0.493
Jensen-Shannon Divergence (↓)	0.315
Continuous ℓ_1 Variation (↓)	0.216
Macro Top-1 Accuracy (↑)	0.410
Evaluation samples	488100

9 Limitations and Ethics

Our analysis emphasizes length, but mobility models should also be audited for geographic, socioeconomic, and temporal subgroup bias. Some repo extensions are currently validated on synthetic or proxy settings and require large real-world studies before deployment conclusions. As always with trajectory data, privacy risk remains material even after anonymization; deployment should include strict governance, aggregation controls, and privacy-preserving training where feasible.

10 Broader Impacts and Safeguards

Potential positive impact. Length-robust trajectory modeling can improve transportation planning, demand forecasting, and emergency response routing by reducing short-vs-long trip performance disparity and improving reliability under perturbations.

Potential negative impact. Mobility models can be misused for surveillance, sensitive population monitoring, or unfair allocation decisions if deployed without governance. Even anonymized trajectories can carry re-identification risk when combined with auxiliary data.

Safeguards in this release. We provide a reproducibility-oriented research release (code, scripts, checked-in summary artifacts, and figure/table generation paths) and do not present deployment claims. We explicitly document robustness failure modes, encourage subgroup auditing before real-world use, and recommend access controls, aggregation thresholds, and privacy-preserving training for operational settings.

11 Conclusion

This work delivers an evaluation-first analysis of length sensitivity in trajectory foundation modeling. Using TrajectoryFM-HMT as an instrumented backbone, we introduce LBTL, LSG, and Length-RobustBench, and provide reproducible evidence that length materially affects behavior even when aggregate performance is strong.

A Appendix: Reproducibility Notes

We ensure high-fidelity reproducibility of our entire suite.

Table 6: External assets used in this work and their licenses/terms.

Asset	Source	License/terms	How used in this work
WorldTrace dataset	Hugging Face: OpenTrace/ WorldTrace	ODbL 1.0	Pretraining/evaluation source; the paper reports fixed local-subset results
H3 geospatial indexing	Uber H3 (`https://h3geo.org`)	Apache-2.0	Multi-resolution tokenization backend
OpenStreetMap Context	OSM data (`https://www.openstreetmap.org`)	ODbL 1.0	Rich context features via OSM index pipeline
UniTraj reference code/checkpoint	`https://github.com/vita-epfl/UniTraj`	AGPL-3.0	External baseline evaluation under shared wrapper

Table 7: UniTraj-style clean evaluation (meters).

Model	Rec MAE	Rec RMSE	Pred MAE	Pred RMSE
External UniTraj	17.16	47.15	77.92	159.92
HMT (regression)	5.37	10.51	15.85	35.52
Relative gain (%)	-68.7	-77.7	-79.7	-77.8

A.1 Licenses and terms for external assets

A.2 Moved Main-Text Tables

Clean UniTraj-style meter-scale comparisons are moved to Appendix Table 7.

A.3 Compute resource budgets

A.4 Checkpoint release artifact

To keep repository history lightweight and avoid Git LFS quota coupling, the primary evaluation checkpoint is distributed as a GitHub Release asset: `https://github.com/Brayden-Zhang/ECE324-Mobility-Model-Project/releases/tag/v1.0-checkpoint`. The direct download URL used by the evaluation scripts is: `https://github.com/Brayden-Zhang/ECE324-Mobility-Model-Project/releases/download/v1.0-checkpoint/hmt_worldtrace_stage1_step_25000.pt`.

B Scope Change Statement

No scope change is declared relative to the current project direction. If scope changes are made later, this section will describe what changed and why.

Table 8: Requested SLURM resource budgets for the main training and evaluation entrypoints used by the paper artifact pipeline. These are scheduler requests encoded in the repository scripts, not measured runtime traces.

Stage	GPU	CPUs	RAM	Wall-time
HMT full pretraining	1x H100	8	64G	24:00:00
HMT macro stage-2	1x H100	8	64G	24:00:00
Full benchmark sweep	1x H100	8	32G	08:00:00
Downstream task suite	CPU-only	–	–	–
External UniTraj train	1x H100	8	32G	08:00:00
External UniTraj eval	1x H100	8	32G	04:00:00
UniTraj CSV comparison	CPU-only	2	4G	00:10:00

NeurIPS Paper Checklist

1. **Claims Question:** Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?
   Answer: Yes.
   Justification: The abstract/introduction claims (LBTL, LSG, LengthRobustBench; code-grounded evaluation framing) are aligned with implemented components, scripts, and reported results in the main text and appendix.
2. **Limitations Question:** Does the paper discuss the limitations of the work performed by the authors?
   Answer: Yes.
   Justification: Section *Limitations and Ethics* explicitly discusses subgroup-auditing gaps, synthetic/proxy validation limits, and privacy risks/governance considerations.
3. **Theory Assumptions and Proofs Question:** For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?
   Answer: Not applicable.
   Justification: The paper is empirical/implementation-focused and does not present formal theorem-proof contributions.
4. **Experimental Result Reproducibility Question:** Does the paper fully disclose all information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?
   Answer: Yes.
   Justification: The paper and repo provide dataset/checkpoint choices, objective definitions, evaluation protocols, and runnable scripts/commands for the main results; Appendix reproducibility notes and README map key entrypoints.
5. **Open Access to Data and Code Question:** Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?
   Answer: Yes.
   Justification: The paper and repository provide the code paths, SLURM entrypoints, checked-in summary artifacts, and figure/table generation scripts needed to regenerate the manuscript-facing outputs from preserved report files; large checkpoints and cache directories remain external to Git.
6. **Experimental Setting/Details Question:** Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?
   Answer: Yes.
   Justification: The manuscript reports dataset subset, checkpoint protocol, stratification thresholds, metrics, objective decomposition/weights, and named scripts for each benchmark component; additional operational details are in appendix/repo commands.
7. **Experiment Statistical Significance Question:** Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?
   Answer: No.

Justification: The current version reports deterministic point estimates from fixed checkpoints and fixed evaluation splits. We explicitly avoid unsupported repeated-run variance claims and treat additional multi-seed significance analysis as future work.

8. **Experiments Compute Resources Question:** For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?
Answer: Yes.
Justification: Appendix Table 8 reports the requested GPU/CPU, memory, and wall-time budgets encoded in the main SLURM entrypoints, and the repository includes those executable scripts directly.

9. **Code Of Ethics Question:** Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?
Answer: Yes.
Justification: The work is an offline modeling/evaluation study with explicit privacy and responsible-use discussion, and no stated deviations from NeurIPS ethics requirements.

10. **Broader Impacts Question:** Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?
Answer: Yes.
Justification: The paper now includes a dedicated *Broader Impacts and Safeguards* section that explicitly describes positive impacts (planning/forecasting/reliability benefits), negative impacts (surveillance, unfair allocation, re-identification risk), and mitigation guidance.

11. **Safeguards Question:** Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse?
Answer: Yes.
Justification: The paper now includes an explicit safeguards paragraph describing release posture, documented failure modes, and recommended governance controls (access controls, aggregation thresholds, privacy-preserving deployment practices).

12. **Licenses for existing assets Question:** Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?
Answer: Yes.
Justification: The appendix now includes a structured external-assets license table (*Licenses and terms for external assets*, Table 6) listing sources, license/terms, and usage context for each asset.

13. **New Assets Question:** Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?
Answer: Not applicable.
Justification: The submission's primary contribution is methodological/evaluative framing and analysis rather than release of a new standalone dataset or model asset package.

14. **Crowdsourcing and Research with Human Subjects Question:** For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?
Answer: Not applicable.
Justification: The paper reports no crowdsourcing experiments or prospective human-subject data collection.

15. **Institutional Review Board (IRB) Approvals or Equivalent for Research with Human Subjects Question:** Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether IRB approvals (or equivalent) were obtained?
Answer: Not applicable.
Justification: No human-subject intervention study is conducted in this work.

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